

# Disinformation analysis YouTube

Maggie Mazza, Casper Polhaar

2026-05-11

## Setup

```
#load neccessary packages
library(tidyverse)
library(tinytex)
library(tidytext)
library(knitr)
library(corr)

# Read the datasets
# (Ensure these files are in your working directory)
video_meta <- read_csv("youtube_haitian_disinformation_video_meta.csv")
comment_meta <- read_csv("youtube_haitian_disinformation_comment_reply_metadata.csv")

# View the main dataset
glimpse(video_meta)
```

```
## Rows: 23
## Columns: 13
## $ ...1      <dbl> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ video_id  <chr> "OVS99PXbbw", "39voRIZciVI", "4IGd0BrrXoI", "f1RF50QrU~
## $ title     <chr> "Why Trump keeps pushing debunked claim about Haitian i~
## $ description <chr> "University of Virginia Center for Politics director La~
## $ date      <dtm> 2024-09-14 22:19:34, 2024-09-15 17:50:19, 2024-09-11 0~
## $ tags      <chr> "[ 'politics' 'us news' 'immigration' 'springfield ohio'~
## $ views     <dbl> 312222, 320834, 1858436, 43414, 236924, 1342405, 261990~
## $ likes     <dbl> 4327, 5938, 27856, 546, 5363, 35919, 4575, 5830, 8761, ~
## $ dislikes  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ comments  <chr> "[ 'What presidential candidate continues rhetoric that ~
## $ channel_title <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Fo~
## $ transcript <chr> "[ { 'duration': 1.335, 'start': 0.133, 'text': 'The mayo~
## $ transcript_str <chr> "The mayor of Springfield, Ohio, issued a public plea s~
```

## Data preparation and comment counting

```
# Count comments per video
comment_counts <- comment_meta %>%
  group_by(video_id) %>%
```

```

summarise(comment_count = n(), .groups = 'drop')

# Merge, clean, and prep the dataset
video_prep <- video_meta %>%
  # Merge the comment counts
  left_join(comment_counts, by = "video_id") %>%
  # If a video had no comments in the comment file, replace NA with 0
  mutate(comment_count = replace_na(comment_count, 0)) %>%
  # Remove irrelevant variables
  select(-any_of(c("date", "tags", "comments", "transcript", "transcript_str"))) %>%
  # Create the dichotomous variable (filled with NA for now so you can code it)
  mutate(video_debunking = NA_real_)

# View the prepped data
head(video_prep)

```

```

## # A tibble: 6 x 10
##   ...1 video_id  title      description  views likes dislikes channel_title
##   <dbl> <chr>      <chr>      <chr>      <dbl> <dbl> <lgl>      <chr>
## 1     0 OVS99PXbbw Why Trump k~ "Universit~ 3.12e5  4327 NA        <NA>
## 2     1 39voRIZciVI Springfield~ "In a one~ 3.21e5  5938 NA        <NA>
## 3     2 4IGd0BrrXoI Trump, Harr~ "Former Pr~ 1.86e6  27856 NA        <NA>
## 4     3 flRF50QrURI Trump ampli~ "Former Pr~ 4.34e4   546 NA        <NA>
## 5     4 GTmdOGut4Gw Springfield~ "A Springf~ 2.37e5  5363 NA        <NA>
## 6     5 L5Gq1Tc63UE Springfield~ "Springfie~ 1.34e6  35919 NA        <NA>
## # i 2 more variables: comment_count <int>, video_debunking <dbl>

```

## Coding the video\_debunking variable

```

# 1. Manually list the exact video IDs that are DEBUNKING
# (Replace these placeholders with your actual video IDs)
debunking_videos <- c("xc0IkVmiN8o", "flRF50QrURI", "6W0mzgcDNvY", "eytlX20TKPs", "Ru34gTyHKlw", "OVS99")

# 2. Apply the manual coding and create the factor
video_final <- video_prep %>%
  mutate(
    # First, create the numeric version for your correlation analysis
    video_debunking_num = ifelse(video_id %in% debunking_videos, 0, 1),

    # Next, create the factor version for your tables and summaries
    video_debunking_factor = factor(video_debunking_num,
                                     levels = c(0, 1),
                                     labels = c("debunking", "not debunking"))
  )

# View a quick summary to make sure your factor applied correctly
summary(video_final$video_debunking_factor)

```

```

##   debunking not debunking
##         14             9

```

## Summarising engagement

```
# Use the factor version so the table reads "debunking" / "not debunking"
overview_table <- video_final %>%
  select(title, video_debunking_factor, views, likes, comment_count)

kable(overview_table,
      caption = "Overview of Video Engagement",
      format.args = list(big.mark = ","))
```

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 1: Overview of Video Engagement

| title                                                                                           | video_debunking_factor | views     | likes  | comment_count |
|-------------------------------------------------------------------------------------------------|------------------------|-----------|--------|---------------|
| Why Trump keeps pushing debunked claim about Haitian immigrants                                 | debunking              | 312,222   | 4,327  | 5,132         |
| Springfield mayor responds to JD Vance's claims about Haitian immigrants                        | not debunking          | 320,834   | 5,938  | 4,070         |
| Trump, Harris face off at the ABC News Presidential Debate                                      | not debunking          | 1,858,436 | 27,856 | 10,803        |
| Trump amplifies false claims about Haitian immigrants eating pets                               | debunking              | 43,414    | 546    | 820           |
| Springfield, Ohio, man reported Haitian migrants snatching geese out of park just 2 weeks ago   | not debunking          | 236,924   | 5,363  | 1,670         |
| Springfield resident describes impact of Haitian migrants on community                          | not debunking          | 1,342,405 | 35,919 | 9,085         |
| Springfield neighbors describe immigrant situation; say they believe pet-snatching story        | not debunking          | 261,990   | 4,575  | 3,581         |
| JD Vance exposed for knowingly lying about 'pet-eating' rumors in Springfield, OH               | debunking              | 280,072   | 5,830  | 1,917         |
| 'You think this is the last cup of B.S.?': Steele slams Trump campaign for lying about migrants | debunking              | 442,417   | 8,761  | 2,108         |
| Ohio residents rage at officials over Haitian migrant surge: 'No vacancy'                       | not debunking          | 1,233,287 | 34,778 | 9,843         |
| Trump spreads more lies about Springfield, Ohio                                                 | debunking              | 297,249   | 3,370  | 4,536         |
| Springfield, Ohio resident: This is a dystopian nightmare                                       | not debunking          | 453,853   | 10,533 | 144           |
| Haitian community in Springfield, Ohio, under threat after debunked comments                    | debunking              | 36,872    | 459    | 916           |
| Reporter Presses JD Vance About 'Debunked False Claims' About Haitians In Springfield, Ohio     | debunking              | 82,768    | 2,238  | 926           |
| Criminal charges filed by Haitian nonprofit against Trump and JD Vance                          | debunking              | 44,919    | 1,496  | 1,290         |
| 'NO DOGS OR CATS' Haitian immigrant pleads with Trump and Springfield, Ohio residents           | debunking              | 54,222    | 809    | 1,349         |

| title                                                                                       | video_debunking_factor | views   | likes  | comment_count |
|---------------------------------------------------------------------------------------------|------------------------|---------|--------|---------------|
| Missing cat at center of Springfield conspiracy theory found hiding in basement             | debunking              | 332,473 | 5,631  | 2,270         |
| Trump falsely claims Haitian migrants are eating pets in Ohio                               | debunking              | 88,241  | 986    | 1,974         |
| JD Vance claims on Ohio's Haitian immigrants eating pets debunked                           | debunking              | 86,482  | 608    | 1,368         |
| Dana Bash and JD Vance clash over baseless claims about Haitian immigrants (Part 1/2)       | debunking              | 838,589 | 12,164 | 1,536         |
| JD Vance reacts to Springfield father saying he politicized his son's death (Part 2/2)      | not debunking          | 379,162 | 6,603  | 101           |
| JD Vance defends amplifying false claims about immigrants                                   | debunking              | 195,846 | 3,467  | 3,969         |
| Residents in Springfield, Ohio react to accusations of immigrants abducting and eating pets | not debunking          | 243,855 | 1,296  | 1,617         |

## Analysing the correlation

```
# Use the numeric version so the correlation math works
correlation_data <- video_final %>%
  select(video_debunking_num, views, likes, comment_count) %>%
  drop_na()
```

```
correlation_results <- correlation_data %>%
  correlate(method = "spearman", diagonal = 1)
```

```
## Correlation computed with
## * Method: 'spearman'
## * Missing treated using: 'pairwise.complete.obs'
```

```
kable(correlation_results,
      digits = 3,
      caption = "Spearman Correlation Matrix")
```

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

```
## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")
```

Table 2: Spearman Correlation Matrix

| term                | video_debunking_num | views | likes | comment_count |
|---------------------|---------------------|-------|-------|---------------|
| video_debunking_num | 1.000               | 0.524 | 0.551 | 0.215         |
| views               | 0.524               | 1.000 | 0.941 | 0.534         |
| likes               | 0.551               | 0.941 | 1.000 | 0.462         |
| comment_count       | 0.215               | 0.534 | 0.462 | 1.000         |